Visual cues in sound change: A cross-modal perceptual account for the typological rarity of labial palatalization.

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Background

Listener-based sound change

- (Mis-)perception of acoustically similar sounds contributes to diachronic sound changes (Ohala, 1981, 1993; Blevins, 2004).

Visual cues in speech perception

- Enhance speech intelligibility (Sumby and Pollack, 1954)
- Alter linguistic percept when presented with conflicting auditory cues (McGurk & McDonald, 1976)
- Visually perceived lip movements activate brain regions responsible for auditory perception (Calvert et al., 1997).

Visual cues in sound change: Understudied

- Unrounded variants of CAUGHT are more likely to be misperceived as COT for Northern Cities-shifted listeners (Havenhill, 2018; Havenhill & Do, 2018).
- Variable interdentality of /θ/ vs. visually stable /f/ contributes to typological asymmetry favoring /θ/ > /f/ changes (McGuire & Babel, 2012).

The typological asymmetry of palatalization

- Palatalization of labials:
  - Extremely rare across languages (Bhat 1978, Kochetov 2011)
  - Limited to secondary but not full palatalization. Change from /pj/ to /tɕ/ is attested in only 2-3 lgs (Bateman, 2010)
- Accounts for labial palatalization in Setswana:
  - Direct sound change due to acoustic similarity (Ohala, 1978)?

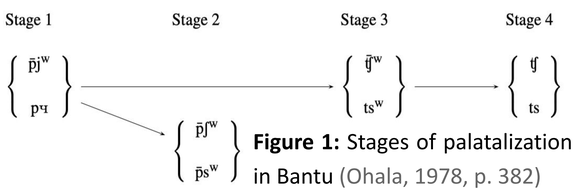

Figure 1: Stages of palatalization in Bantu (Ohala, 1978, p. 382)

  - Indirect change via telescoping: Hardening of fricative noise and subsequent deletion of labials (Bateman, 2010)?

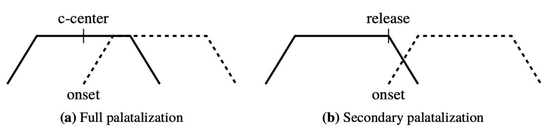

Figure 2: Full palatalization is motivated by gestural blending for lingual but not labial consonants (from Bateman 2007). Solid line: consonantal gesture. Dashed line: vocalic gesture.

The current study

- To what extent do visual cues contribute to or inhibit misperception-based sound change?
- Given diminished auditory place cues, why is [pʲ] seemingly not misperceived as [tʃ/tɕ]?
- Hypothesis: **Visible lip closure offers additional cue to place of articulation (PoA) and inhibits misperception**

The Perception Experiment

Study Design

- 3 conditions: Audio-only, Audiovisual Congruent, and Audiovisual Incongruent
- **Audio stimuli: Interpolation of natural tokens produced by female Korean speaker** in Matlab.
  - All acoustic cues were interpolated in 5 steps.
  - 3 sets of continua: /pʰV-tɕV/, /tʰV-tɕV/, /kʰV-tɕV/ with vowels /i, a, o, u, ɛ, ʌ/ and glide /∅, j/
- **Visual stimuli**
  - Congruent: original pairing of audio and video
  - Incongruent: swapping lip-opening and lip-closure
- Fillers: /m, n, h, j, w/ + /i, a, o, u, ɛ, ʌ/

Procedure

- 2-alternative forced choice task through Prolific
- Fixation cross → Audio/Audio+video → Response
- Reaction time and keyboard response collected
- Participants: 75 Korean speakers (42 F/31 M/2 other)

Condition	Audio	Visual cue	Choices
Audio Only	5-step continuua for each onset-vowel combination	NA	stops vs. affricate
Congruent		Labial: Lip-closure Non-labial: Lip-opening	
Incongruent		Labial: Lip-opening Non-labial: Lip-closure	

Table 1: The experiment design

Analysis and Results

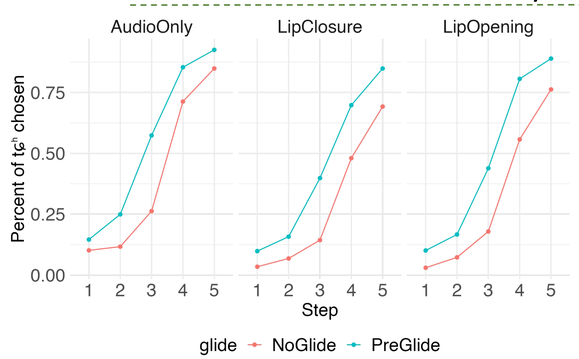

Figure 3: The effect of glide on perceptual boundaries

Item	Estimate	Std. Err	z value	Pr(>|z|)
(Intercept)	-4.22	0.14	-29.65	<0.001 ***
Step	1.18	0.02	54.49	<0.001 ***
Glide (PreGlide)	1.02	0.05	20.76	<0.001 ***
Condition (Congruent)	-0.89	0.16	-5.64	<0.001 ***
Condition (Incongruent)	-0.43	0.16	-2.66	<0.01 **
Onset (ph)	-0.25	0.06	-4.30	<0.001 ***
Onset (th)	0.27	0.06	4.63	<0.001 ***

Table 2: Logistic Mixed-Effects Regression

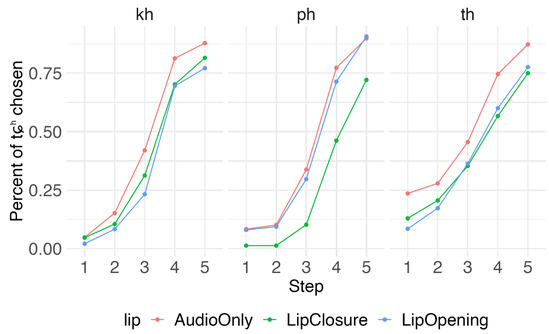

Figure 4: Visual cues shift perceptual boundaries

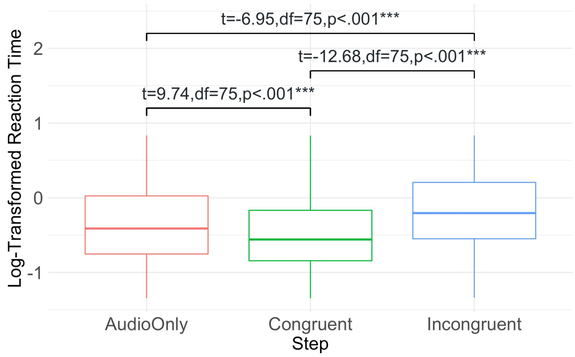

Figure 5: Reaction time in three conditions

Analysis of choices and reaction time

- Responses with RT 2-SE away from the global mean are excluded.
- Focus: Perceptual boundary shifts and cognitive load.

Acoustic cues induce misperception

- In Audio-only condition, all three stops were liable to be perceived as palatals.
- Glide /j/ (Figure 3, β=1.02, SE=0.05, p<.001***) and high vowels (t=12.73, df=11613, p<.001***) favors misperception.

Lip closure inhibits the misperception

- Figure 4: Perceptual boundary shifts towards affricate.

Lip opening favors the misperception

- Figure 4: Perceptual boundary shifts towards /tɕ/

Congruent cues expedite speech processing

- Figure 5: Congruent (61.69ms) vs. Audio Only (70.22ms)

Incongruent cues slows speech processing down

- Figure 5: Incongruent (76.44ms) vs. Audio Only (70.22ms)

Discussion

- Both acoustic and visual cues significantly influence perception of stop-affricate place contrast in Korean.
- Perception of cues in both dimensions is line with typological patterns.
  - High vowels, glide /j/, and lip opening gesture favor synchronic misperception of stops as affricates.
  - Low vowels, absence of glide, and visible lip closure gesture disfavor such misperception.
- Audiovisual congruence improves identification of phonological category.
- Incongruence for onsets originally with lip opening gesture, increases reaction time and cognitive load.
- Future directions:
  - How do acoustic place cues (F2, burst/fricative noise) trade between each other and with visual cues?
  - To what extent do speakers optimize their productions for visual perceptibility? (see 2pSC22, this session)
  - How do visual cues of the same feature/category differ? (see 3aSC36, tomorrow)

Conclusion

- Full labial palatalization is cross-linguistically rare, while coronals and dorsals are frequently palatalized.
- Acoustically palatalized stimuli are less likely to be perceived as [tɕ] when shown with visible lip closure.
- This study suggests that both visual cues (i.e., lip closure) *and* articulatory factors contribute to the asymmetry.
- Additional analyses also support previous findings about the role of audiovisual congruence in speech processing.

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